

Amirhossein Abaskohi

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in amirhossein-abaskohi 🔄 AmirAbaskohi 🎓 Google Scholar

Education

University of British Columbia — Ph.D. in Computer Science

Vancouver, Canada

- **GPA:** A+
- **Supervisors:** Giuseppe Carenini, Peter West

Sep 2025 — Present

University of British Columbia — M.Sc. in Computer Science

Vancouver, Canada

- **GPA:** A+
- **Supervisor:** Giuseppe Carenini
- **Thesis:** Multimodal understanding of long documents : from topic modeling to question answering [🔗](#)

Sep 2023 — Aug 2025

University of Tehran — B.Sc. in Computer Engineering

Tehran, Iran

- **CGPA:** 3.8/4 (Last Two Years GPA: 4/4)
- Ranked Top 10% in class

Sep 2018 — Apr 2023

Research Interests

- Multimodal Reasoning
- Long Document Understanding
- Deep Research
- NLP for Commonsense Reasoning
- Human-Centered AI
- Interpretable AI

Skills

ML/AI: PyTorch, TensorFlow, HuggingFace, Reinforcement Learning (RL), RLHF, RAG, ReAct, Chain-of-Thought (CoT), Tool Calling, OpenAI/Anthropic APIs, CUDA, Scikit-learn, spaCy, LangChain

Programming: Python, C/C++, Java, C#, SQL, JavaScript, MATLAB

Software/DevOps: Git, Docker, Flask, Django, CI/CD, Airflow, RabbitMQ, AWS Cloud

Databases & Analytics: PostgreSQL, MySQL, MongoDB, Spark, Elasticsearch, Grafana, PowerBI

Publications

2026

- **SeKV: Resolution-Adaptive KV Cache with Hierarchical Semantic Memory for Long-Context LLM Inference** [🔗](#)
A. Abaskohi, G. Carenini, P. West, Y. He
arXiv.
- **MCompassRAG: Topic Metadata as a Semantic Compass for Paragraph-Level Retrieval** [🔗](#)
A. Abaskohi, R. Li, G. Cimino, P. West, G. Carenini, I. H. Laradji
arXiv.
- **SproutRAG: Attention-Guided Tree Search with Progressive Embeddings for Long-Document RAG** [🔗](#)
A. Abaskohi, I. H. Laradji, P. West, G. Carenini
arXiv.
- **Dual-Confidence Contrastive Decoding for Retrieval-Augmented Generation** [🔗](#)
R. Li, Md. T. Islam Khondaker, **A. Abaskohi**, G. Murray, G. Carenini, I. H. Laradji
arXiv.
- **UnpredictaBench: A Benchmark for Evaluating Distributional Randomness in LLMs** [🔗](#)
A. Abaskohi*, A. Dabiriaghdam*, L. Luo, ED. Wang, G. Carenini, P. West
arXiv.
- **ReVision: Scaling Computer-Use Agents via Temporal Visual Redundancy Reduction** [🔗](#)
A. Abaskohi, Y. He, P. West, G. Carenini, P. Chawla, V. Vineet
arXiv.

- **Improving Neural Topic Modeling with Semantically-Grounded Soft Label Distributions** [↗](#)
R. Li, **A. Abaskohi**, C. Li, G. Murray, G. Carenini
ICML 2026.
- **DRBench: A Realistic Benchmark for Enterprise Deep Research** [↗](#)
A. Abaskohi, T. Chen, M. Muñoz-Mármol, C. Fox, A. Varshini Ramesh, É. Marcotte, X. H. Lù, N. Chapados, S. Gella, P. West, G. Carenini, C. Pal, A. Drouin, I. H. Laradji.
ICLR 2026.
- **StarFlow: Generating Structured Workflow Outputs From Sketch Images** [↗](#)
P. Bechard, C. Wang, **A. Abaskohi**, J. Rodriguez, C. Pal, D. Vazquez, S. Gella, S. Rajeswar, P. Taslakian.
EACL 2026.

2025

- **CEMTM: Contextual Embedding-based Multimodal Topic Modeling** [↗](#)
A. Abaskohi, R. Li, C. Li, S. Joty, G. Carenini.
EMNLP 2025.
- **ChartGaze: Enhancing Chart Understanding in LLMs with Eye-Tracking Guided Attention Refinement** [↗](#)
A. Salamatian, **A. Abaskohi**, W.-C. Fan, M. R. I. Hossain, L. Sigal, G. Carenini.
EMNLP 2025.
- **AgentAda: Skill-Adaptive Data Analytics for Tailored Insight Discovery** [↗](#)
A. Abaskohi, A. V. Ramesh, S. Nanisetty, C. Goel, D. Vazquez, C. Pal, S. Gella, G. Carenini, I. H. Laradji.
REALM@ACL 2025 (Spotlight).
- **FM²DS: Few-Shot Multimodal Multihop Data Synthesis with Knowledge Distillation for Question Answering** [↗](#)
A. Abaskohi, S. Gella, G. Carenini, I. H. Laradji.
EMNLP Findings 2025.

2024

- **BigDocs: An Open and Permissively-Licensed Dataset for Training Multimodal Models on Document and Code Tasks** [↗](#)
J. A. Rodriguez, X. Jian, S. S. Panigrahi, T. Zhang, A. Feizi, A. Puri, A. Kalkunte, F. Savard (Equal 1st), **A. Abaskohi** (Equal 2nd), et al.
ICLR 2025 & RBFM@NeurIPS 2024.
- **InsightBench: Evaluating Business Analytics Agents Through Multi-Step Insight Generation** [↗](#)
G. Sahu, A. Puri, J. Rodriguez, **A. Abaskohi**, et al.
ICLR 2025.
- **Benchmarking Large Language Models for Persian: A Preliminary Study Focusing on ChatGPT** [↗](#)
A. Abaskohi, S. Baruni, M. Masoudi, et al.
LREC-COLING 2024.
- **Beyond Words: A Multimodal and Multilingual Exploration of Persuasion in Memes** [↗](#)
A. Abaskohi^{*}, A. Dabiriaghdam^{*}, L. Wang, G. Carenini.
SemEval@NAACL 2024.
- **Can LLMs be Lateral Thinkers?** [↗](#)
P. Sadeghi^{*}, **A. Abaskohi**^{*}, Y. Yaghoobzadeh.
SemEval@NAACL 2024.

2023

- **Paraphrasing-Guided Data Augmentation for Contrastive Prompt-Based Few-Shot Fine-Tuning** [↗](#)
A. Abaskohi, Y. Yaghoobzadeh, S. Rothe.
ACL 2023.
- **PEACH: Pre-Training Sequence-to-Sequence Multilingual Models for Translation with Semi-Supervised Pseudo-Parallel Document Generation** [↗](#)
A. Abaskohi^{*}, A. Salemi^{*}, S. Tavakoli, Y. Yaghoobzadeh, A. Shakeri.
LoResMT@EACL 2023.
- **A Large-Scale Analysis of Persian Tweets Regarding Covid-19 Vaccination** [↗](#)
T. ShabaniMirzaei, H. Chamani, **A. Abaskohi**, Z. Sourati, B. Bahrak.
Social Network Analysis and Mining (Springer).

2022

- **Persian Emotion Detection using ParsBERT and Imbalanced Data Handling Approaches** [↗](#)
A. Abaskohi, N. Sabri, B. Bahrak.
arXiv preprint.
- **A Comparative Analysis of Sarcasm Detection Using Generative-based and Mutation-based Data Augmentation** [↗](#)
A. Abaskohi, A. Rasouli*, T. Zeraati*, B. Bahrak.
SemEval@NAACL 2022.
- **Automatic Speech Recognition for Speech Assessment of Persian Preschool Children** [↗](#)
A. Abaskohi, F. Mortazavi, H. Moradi.
arXiv.

Volunteering

- **Publicity Chair, Future of NLP Workshop** [↗](#) — Oct 2024–Dec 2024
- **AI/ML Mentor, Summer School** [↗](#) — Summer 2021
- **Content Creator, Medium** [↗](#) — Jul 2021–Present
- **Content Creator, YouTube** [↗](#) — Jul 2021–Present
- **Backend API Developer, Cognitive Tests for Preschool Children** — Nov 2020–Feb 2021

Honors & Awards

President's Academic Excellence Initiative PhD Award (PAEIPA) Recognizes outstanding academic excellence among UBC doctoral students, awarded upon admission to the PhD program.	<i>Aug 2025</i>
Graduate Support Initiative (GSI) Award Merit-based award for research excellence and academic achievement among UBC graduate students.	<i>Aug 2025</i>
Spotlight Paper (AgentAda), REALM@ACL Selected as a Spotlight presentation at the ACL REALM 2025 workshop for its novelty and high reviewer ratings.	<i>May 2025</i>
Mitacs Fellowship (Project IT41557) Competitive national fellowship supporting academic–industry collaboration during research at ServiceNow Research.	<i>Jun 2024</i>
UBC International Tuition Award Awarded to outstanding international students demonstrating strong research potential and academic excellence.	<i>Apr 2023</i>
UBC CS Strife Award Granted for exemplary performance and contributions within the Department of Computer Science.	<i>Apr 2023</i>
Best Undergraduate Project Award Recognized as the top senior-year capstone project at the University of Tehran for innovation and technical depth.	<i>Sep 2022</i>
Supporter Foundation of University of Tehran Scholarship Merit-based scholarship awarded to top-performing students in the Computer Engineering program.	<i>Sep 2020</i>
Admission to University of Tehran Secured admission among the top 1% nationwide (National rank 1125 / 144,437; Regional rank 46) in Iran's national entrance exam.	<i>Sep 2018</i>